

Image Enhancement:-

- (i) Contrast Improvement (ii) Noise cleaning (iii) Image Sharpening

Contrast may be compromised — Contrast stretching expands the

- (a) Improper illumination range of intensity levels in an image so that it spans the ideal full intensity range of the device.

(b) Aperture

(c) Shutter speed

(d) Lack of dynamic range in imaging sensor.

shape of and range of the intensity histogram reflects a lot. Poor contrast means histogram shows that dynamic range of intensities is confined in a small range.

Linear stretching:-

$m \rightarrow$ Input intensity

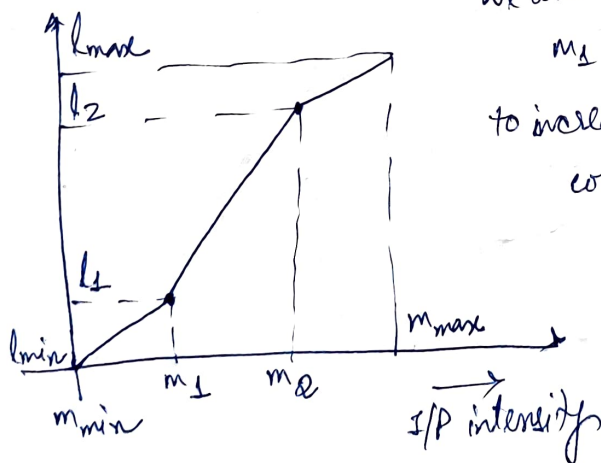
$$T(m) = \frac{[m - [m_{\min}, m_{\max}]]}{[l_{\min}, l_{\max}]} \rightarrow \text{Input range} \rightarrow \text{Allowable range}$$

for any $m_2 > m_1$ monotonically increasing

$T(m_2) \geq T(m_1) \rightarrow$ cannot properly map reversely

$$T(m) = \frac{l_{\max} - l_{\min}}{m_{\max} - m_{\min}} (m - m_{\min}) + l_{\min}$$

Piecewise Linear Stretching:-



We want to enhance m_1 to m_2 part to increase range of coverage

2. image

① If $m \leq m_1$, $\frac{l_1 - l_{min}}{m_1 - m_{min}} (m - m_{min}) + l_{min}$

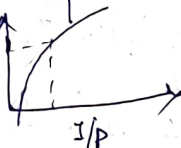
② If $m \geq m_1$ and $m \leq m_2$, $\frac{l_2 - l_1}{m_2 - m_1} (m - m_1) + l_1$

③ If $m \geq m_2$, $\frac{l_{max} - l_2}{m_{max} - m_2} (m - m_2) + l_2$

Logarithmic Stretching

Majority pixels are occupying a narrow lower intensity range $\rightarrow$ Dark image
to enhance contrast, lower intensity values are to be more stretched at the cost of compressed stretching of higher intensity value.

$$T(m) = (l_{max} - l_{min}) \cdot \frac{\log(m - m_{min} + 1)}{\log(m_{max} - m_{min} + 1)} + l_{min}$$

If any captured image follows exponential characteristics, then apply logarithmic function, O/P 

Exponential Stretching

A change of brightness δB at a point with its neighbourhood avg intensity B .

$\frac{\delta B}{B}$ is weber's ratio $\rightarrow$ constant

To make the change δB noticeable increase δB with increase in B ,

$$[m_{min}, m_{max}] \rightarrow [l_{min}, l_{max}]$$

no. of levels, $K = m_{max} - m_{min} + 1$

$$l_{i+1} = l_i + \delta B$$

Weber's ratio = $\frac{l_{i+1}}{l_i} = \frac{l_2}{l_1} = \frac{l_3}{l_2} = \dots = \frac{l_k}{l_{k-1}} = C$ [constant]

$$I_k = c I_{k-1} = c^2 I_{k-2} = \dots = c^{k-1} I$$

$$c = \left(\frac{I_k}{I_1} \right)^{\frac{1}{k-1}} = \left(\frac{I_{\max}}{I_{\min}} \right)^{\frac{1}{m_{\max} - m_{\min}}}$$

$$T(m) = I_{\min} \cdot c^{(m - m_{\min})}$$

$$= I_{\min} \left(\frac{I_{\max}}{I_{\min}} \right)^{\frac{m - m_{\min}}{m_{\max} - m_{\min}}}$$

$$\begin{array}{l} m_{\min} \rightarrow I_{\min} \\ m_{\min+1} \rightarrow c I_{\min} \\ \vdots \\ m \rightarrow c^{m - m_{\min}} \cdot I_{\min} \end{array}$$

Histogram Equalization:-

Let, initially the variable r denote the intensities of an image to be processed, assuming r is in the range $[0, L-1]$

Let $P_r(r)$ be Pdf of intensity value r .

And random variable $s = T(r)$ say $P_r(r) \rightarrow \text{known}$
 $T(r) \rightarrow \text{known}$

Let, $P_s(s)$ be PDFs of intensity value s .

There are L levels, so $(L-1)$ intervals we have to find $P_s(s)$

$$T(r) = (L-1) \int_0^r P_r(w) dw \quad \begin{array}{l} \text{Input} \rightarrow [0, L-1] \\ T(r) \rightarrow [0, L-1] \end{array}$$

cdf (cumulative distribution function)

$$T(r_1) = (L-1) \int_0^{r_1} P_r(w) dw$$

If $r_2 > r_1$ then $T(r_1) < T(r_2)$

$$P_s(s) = P_r(r) \left| \frac{dr}{ds} \right|$$

$$s = T(r) = (L-1) \int_0^r P_r(w) dw$$

$$\frac{ds}{dr} = (L-1) \frac{d}{dr} \left[\int_0^r P_r(w) dw \right]$$

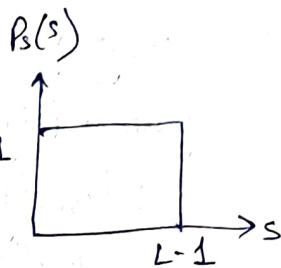
Assuming $P_r(r)$ is continuous and differentiable.

$$\Rightarrow \frac{ds}{dr} = (L-1) P_r(r)$$

$$\Rightarrow P_s(s) = P_r(r) \cdot \frac{1}{(L-1) P_r(r)} \cdot \frac{1}{L-1}$$

$$\Rightarrow P_s(s) = \frac{1}{L-1} \quad 0 \leq s \leq L-1$$

[uniform PDF]



Probability of occurrence of intensity level $r_k, P_r(r_k) = \frac{n_k}{MN}$
 [$n_k \rightarrow$ no. of pixels having intensity r_k]
Histogram Matching / Specification
 $S = T(r) = (L-1) \int_0^r P_r(w) dw$
 Input histogram $\rightarrow$ Target histogram
 $r \quad \quad \quad z$

Another function on z , $G(z) = (L-1) \int_z^Z P_z(v) dv = S$
 [w and v are dummy variable of integration]

$$G(z) > S$$

$$\Rightarrow z > G^{-1}(S) = G^{-1}(T(r))$$

for discrete histogram specification —

① Perform histogram equalization from $P_r(r)$ of the input image and use it to map the intensities in the target image. round the resulting values, s_k to $[0, L-1]$

$P_z(z)$ is given, G in $[0, L-1]$.

Find corresponding value of z_q so that $G(z_q)$ is closest to s_k .
 Store these mappings from s to z .

$$T_1 \quad r \mid s \quad T_2 \quad z \mid s$$

- ② for each value of r , say r_k from T_1 get corresponding s_k
- ③ Search in T_2 for s_k , if single match found get the corresponding z
- if multiple match,
select z as per convention (smallest one)

Local contrast enhancement

$(x, y) \rightarrow$ neighbourhood mean intensity / brightness
global brightness $\rightarrow \bar{f}$

intensity variance over the neighbourhood $\sigma^2(x, y)$

(scaling factor)

$$f'(x, y) = K [f(x, y) - \bar{f}(x, y)] + \bar{f}(x, y)$$

$$K = c \cdot \frac{\bar{f}}{\sigma^2(x, y)}$$

$0 \leq c \leq 1 \rightarrow$ constant

If $\sigma^2(x, y) > \text{threshold}$
 $\Rightarrow f'(x, y) = K(f(x, y) - \bar{f}(x, y) + \bar{f}(x, y))$
else
 $f'(x, y) = f(x, y)$

Histogram equalization produces a transformation function that seeks to generate an output image with a uniform histogram. When automatic enhancement is desired, this is a good approach to consider, as the results are predictable and method is simple to implement.

The method used to generate images that have a specified histogram is called histogram matching or histogram specification.

Filter:-

The name 'filter' is borrowed from frequency domain processing where 'filtering' refers to passing, modifying or rejecting specified frequency components of an image.

Ex:- A filter that passes low frequencies is called a lowpass filter.

The net effect produced by a lowpass filter is to smooth an image by blurring it.

Spatial filtering modifies an image by replacing the value of each pixel by a function of the values of the pixel and its neighbours.

If the operation performed on the image pixels is linear, then the filter is called a linear spatial filter, otherwise it's a nonlinear spatial filter.

A linear spatial filter performs a sum-of-products operation between an image f and a filter kernel, w .

The kernel is an array whose size defines the neighbourhood of operation and whose coefficients determine the nature of the filter.

For a kernel of size $m \times n$, $m = 2a + 1$, $n = 2b + 1$
 $[a, b \geq 0]$

The center coefficient of the kernel, $w(0,0)$ aligns with the pixel at location (x,y) .

So, linear spatial filtering of an image of size $M \times N$ with a kernel of size $m \times n$ is —

$$f'(x,y) = \sum_{i=-a}^a \sum_{j=-b}^b w_{ij} * f(x+ai, y+bj)$$
$$= \sum_{i=0}^{2a} \sum_{j=0}^{2b} w'_{ij} * f(x-a+i, y-b+j)$$

Separable Filter Kernels :-

A 2D-function,

$G(x, y)$ is said to be separable if it can be written as the product of two 1-D functions, $G_1(x)$ and $G_2(y)$

$$\text{i.e. } G(x, y) = G_1(x) G_2(y).$$

A spatial filter kernel is a matrix and a separable kernel is a matrix that can be expressed as the outer product of two vectors,

$$W = v w^T, \text{ where } v \text{ and } w \text{ are vectors of size } (m \times 1) \text{ and } (n \times 1) \text{ respectively.}$$

For an image of size $M \times N$ and kernel of size $m \times n$, it requires ' $m n M N$ ' operations. This is because each pixel in the output image depends on all the coefficients in the filter kernel.

But if kernel is separable then, first convolution $w_1 \star f$ requires ' $m N$ ' operations and the second convolution requires ' $n M N$ ' operations, so total $(m+n) M N$ operations, thus, the computational

$$\text{advantage, } C = \frac{M N m n}{M N (m+n)} = \frac{m n}{m+n}.$$

$$w \star f = (w_1 \star w_2) f = (w_2 \star w_1) f = w_2 \star (w_1 \star f) = (w_1 \star f) \star w_2$$

If rank of product matrix is 1, kernel is separable.

■ How spatial filter kernels are constructed?? —

① formulating filters based on mathematical properties.

(say avg, derivative etc).

② Sampling a 2D spatial function whose shape has a desired interest.

③ spatial filter with a specified frequency response.

SMOOTHING (Low Pass Spatial filter) :-

Smoothing (also called averaging) spatial filters are used to reduce sharp transitions in intensity. Because random noise typically consists of sharp transitions in intensity, an obvious application of smoothing is noise reduction. Smoothing is used to reduce irrelevant detail in an image, where "irrelevant" refers to pixel regions that are small w.r.t. the size of the filter kernel.

Another application is for smoothing the false contours that result from using an insufficient no. of intensity levels in an image. May arise because of digitization process / transmission, quantization & their combinations.

$$g(x, y) = f(x, y) + n(x, y)$$

$$g_i(x, y) = f_i(x, y) + n_i(x, y) \rightarrow \text{Transmitted no. of times}$$

$$\rightarrow \frac{1}{n} \sum_{i=1}^n g_i(x, y) = \frac{1}{n} \sum_{i=1}^n f_i(x, y) + \frac{1}{n} \sum_{i=1}^n n_i(x, y)$$
$$= \frac{1}{n} \times n f(x, y) + \frac{1}{n} \sum_{i=1}^n n_i(x, y)$$

If n is large noise will become 0 (zero mean)

① Box Filter :-

An $m \times n$ box filter is an $m \times n$ array of 1's with a normalizing constant in front, whose value is 1 divided by the sum of the values of the coefficients i.e. $\frac{1}{mn}$.

$$\text{If } m=2k+1, n=2k+1$$

$$\hat{g}(x, y) = \frac{1}{(2k+1)(2k+1)} \sum_{i=-k}^{+k} \sum_{j=-k}^{+k} g(x+i, y+j)$$

The normalization has two purposes —

- ① The avg value of an area of constant intensity would be equal that intensity in the filtered image.
- ② It prevents introducing a bias during filtering. Sum of the intensities in the neighbourhood remains unchanged.

Ex:-

3X3 Kernel

1	1	1
1	1	1
1	1	1

convolve with itself

1	2	3	2	1
2	4	6	4	2
3	6	9	6	3
2	4	6	4	2
1	2	3	2	1

→ 5X5 Kernel
Weighted
Mean
Symmetric

Any Symmetrical kernel with non-negative weights such that the weight value is monotonically ~~decreasing~~ ^{increasing} when it move away from the centre can be used for smoothing. This is weighted mean filter. The filter coefficients depends on the position of the element in the weight, and the filtered value is linear combination neighbourhood intensities. [Mean filter blurs the good edges also. Ordered statistic filter preserves good edges to some extent]

ORDERED STATISTIC FILTER ← (nonlinear spatial filter)

The filter value depends on the position of ordering of intensities of neighbourhood pixels. Smoothing is achieved by replacing the value of the center pixel with the value determined by the ranking result.

Median filters provide excellent noise reduction capabilities for certain types of random noise, with less blurring than linear smoothing filters of same size. It is particularly effective in the presence of impulse noise (salt and pepper noise).

Median Filter

Sort the values of the pixels in the neighbourhood based on intensity, $v_1, v_2, \dots, v_n$ (ordered values)

$$\hat{g} = \frac{1}{\sum w_i} \sum w_i v_i \quad w_i = 1 \text{ if } i = \frac{n}{2} \\ \text{else } w_i = 0$$

$$\hat{g}(x, y) = \frac{1}{\sum \sum w(i, j)} \sum \sum w(i, j) g(x, y)$$

Mid Range Filter - Soft and unaffected tails, mid range values are affected, $w_i = 1$ if $1 \leq i \leq k$ or $n-k+1 \leq i \leq n$

$$\text{else } w_i = 0 \\ \sum w_i = 2k \Rightarrow \frac{1}{2k} \sum w_i v_i$$

Trimmed Mean Filter

opposite of the mid range filter, fat tail,

$$w_i = 1 \text{ if } k \leq i \leq n-k \\ \text{else } w_i = 0, \sum w_i = n - 2k \text{ ignored}$$

Max/Min filter

$$\hat{g}(x, y) = \max \{x_i\} \quad w_n = 1 \\ w_i = 0, i \neq n$$

$$\hat{g}(x, y) = \min \{x_i\} \quad w_1 = 1 \\ \text{else } w_i = 0$$

max followed by min (max-min filter)
min followed by max (min-max filter)

alternatively median filter we can use

helps in
salt and
pepper
noise.

① Gaussian low-pass filter :- (Spatial)

Box filters favor blurring along perpendicular directions, the directionality often produces undesirable results,

Gaussian kernel $\rightarrow$ (i) isotropic (their response is independent of orientation)

(ii) circularly symmetric $\rightarrow$ that are also separable.

Gaussian kernels

have zero mean

$$G(x, y) = K e^{-\frac{x^2 + y^2}{2\sigma^2}} \left[K \text{ is constant} = \frac{1}{2\pi\sigma^2} \right]$$

If $r = \sqrt{x^2 + y^2} \Rightarrow G(x, y) = K e^{-\frac{r^2}{2\sigma^2}}$
 r is the distance from the center to any point on function G .

σ = scale of blur
 beyond $\mu = 3\sigma^2$, coefficient is almost zero.

FREQUENCY DOMAIN

$$f(x, y) \xrightarrow{\text{FFT}} F(u, v) \quad (M \times N)$$

$$\text{DFT} \quad F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} e^{-j2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right)} f(x, y)$$

$$\text{IDFT} \quad f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} e^{j2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right)} F(u, v)$$

$H(u, v) \rightarrow$ frequency domain filter

$$\begin{cases} H(u) = A e^{-\frac{u^2}{2\sigma^2}} \\ h(x) = \sqrt{2\pi} \sigma A e^{-\frac{x^2}{2\sigma^2}} \end{cases}$$

$$f^{-1} \left(H(u, v) F(u, v) \right) \xleftarrow{\text{FFT}} F(u, v) H(u, v)$$

$$f(x, y) \quad h(x, y)$$

$h(x, y) \rightarrow$ impulse response of
 $H(u, v) \rightarrow$ corresponding spatial domain $h(x, y)$

1D Gaussian filter

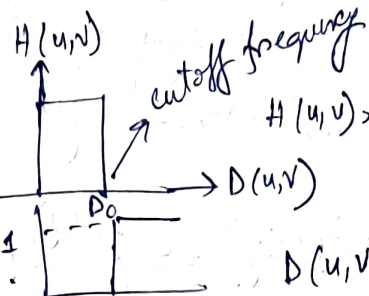
Impulse function

$$\delta(x,y) \xrightarrow{\text{FFT}} \underbrace{F^{-1}(\delta(x,y))}_{1} H(u,v) = H(u,v)$$

Inverse

Frequency Domain filters + (ILPF)

Ideal low Pass filter
Ideal high Pass filter



$$H(u,v) = \begin{cases} 1 & \text{if } D(u,v) \leq D_0 \\ 0 & \text{if } D(u,v) > D_0 \end{cases}$$

$$D(u,v) = \sqrt{(u - P/2)^2 + (v - Q/2)^2}$$

pxs frequency rectangle

ILPF $\Rightarrow H(u,v) = 1 - H_L(u,v) \Rightarrow H(u,v) = \begin{cases} 0 & \text{if } D(u,v) \leq D_0 \\ 1 & \text{if } D(u,v) > D_0 \end{cases}$

Gaussian low Pass filter + (GLPF)

$$H(u,v) = e^{-\frac{D^2(u,v)}{2D_0^2}}$$

Butterworth filter + (BLPF)

$$H(u,v) = \frac{1}{1 + \left(\frac{D(u,v)}{D_0}\right)^{2n}}$$

BLPF function can be controlled to approach the characteristics of the ILPF using higher values of n and the GLPF for lower values of n , while providing a smooth transition in form low to high frequency. We can use BLPF to approach the sharpness of an ILPF function with considerably less ringing.

Gaussian High Pass Filter + (GHPF)

$$H(u,v) = 1 - e^{-\frac{D^2(u,v)}{2D_0^2}}$$

Butterworth High Pass Filter + (BHPF)

$$H(u,v) = \frac{1}{1 + \left(\frac{D_0}{D(u,v)}\right)^{2n}}$$

Transformation funcⁿ

$$\delta(x,y) \xrightarrow{\text{FFT}} F(\delta(x,y)) \xrightarrow{H(u,v)} \underbrace{F(\delta(x,y))}_{1} \cdot H(u,v)$$

$$\xleftarrow{\text{IFFT}} \begin{matrix} u(x,y) \\ \text{(Spatial Domain)} \end{matrix} \quad \begin{matrix} H(u,v) \\ \text{(Freq Domain)} \end{matrix}$$

Image Sharpening

Sharpening highlights transitions in intensity. Image differentiation enhances edges and other discontinuities and de-emphasizes areas with slowly varying intensities.

$$f'(x, y) = \nabla^2 f(x, y) = \frac{\partial^2 f(x, y)}{\partial x^2} + \frac{\partial^2 f(x, y)}{\partial y^2}$$

Taylor Series $\rightarrow$

$$f(x+\Delta x) = f(x) + \frac{\Delta x}{1!} \frac{\partial f(x)}{\partial x} + \dots$$

If $\Delta x = 1$

$$f(x+1) = f(x) + \frac{\partial f(x)}{\partial x} + \dots$$

Ignoring higher terms

$$\Rightarrow f(x+1) = f(x) + \frac{\partial f(x)}{\partial x}$$

$$\Rightarrow \frac{\partial f(x)}{\partial x} = f(x+1) - f(x) \Rightarrow \text{forward difference}$$

$$\text{If } \Delta x = -1 \Rightarrow \frac{\partial f(x)}{\partial x} = f(x) - f(x-1) \Rightarrow \text{backward difference}$$

$$\text{Central Difference} \Rightarrow \frac{f(x+1) - f(x-1)}{2}$$

$$\frac{\partial^2 f(x)}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$$

for first order derivative $\rightarrow$

(i) Zero in areas of constant intensity

(ii) Nonzero at the onset of an intensity ramp or step.

(iii) Along intensity ramp must be, nonzero

for second order derivative $\rightarrow$

(i) zero in areas of constant intensity.

(ii) Must be nonzero at the onset & end of an intensity ramp or step

(iii) Must be zero along the intensity ramps.

$$\nabla^2 f(x, y) = \frac{\partial^2 f(x, y)}{\partial x^2} + \frac{\partial^2 f(x, y)}{\partial y^2}$$

$$= f(x+1, y) + f(x-1, y) - 2f(x, y)$$

$$+ f(x, y+1) + f(x, y-1) - 2f(x, y)$$

$$= f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$

4-neighbours

0	1	0
1	-4	1
0	1	0

8-neighbours

1	1	1
1	-8	1
1	1	1

Spatial Kernel

visually poor
↓ Image crispness
Add -ve of Laplacian
to the original

$$\frac{1}{4} \nabla^2 (f(x, y)) = g_{b4} - f(x, y)$$

$$\frac{1}{8} \nabla^2 (f(x, y)) = g_{b8} - f(x, y)$$

$$f(x, y) = g_{b4} - f(x, y) \quad \text{crispness}$$

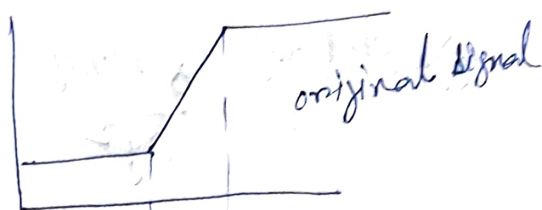
$$\hat{f}_c(x, y) = f(x, y) - g_{b4} + f(x, y) \\ = 2f(x, y) - g_{b4}$$

Unsharp masking:-

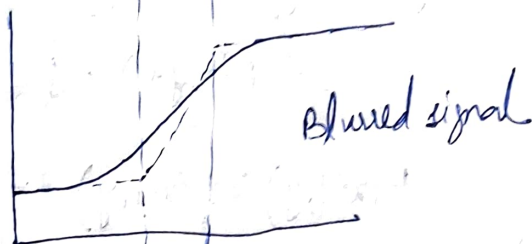
Subtracting an unsharp version of an image from the original image is process to sharpen images. This process is called unsharp masking. The steps are —

- ① Blur the original image.
- ② Subtract the blurred image from the original to generate the mask.
- ③ Add the mask to the original.

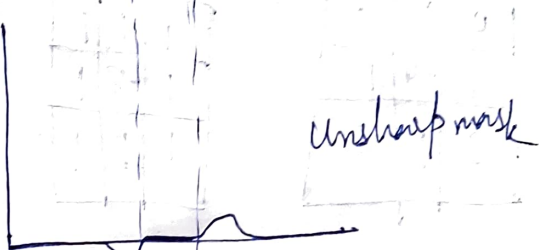
$f(x, y)$



$\bar{f}(x, y)$



$$g_{\text{mask}}(x, y) = f(x, y) - \bar{f}(x, y)$$



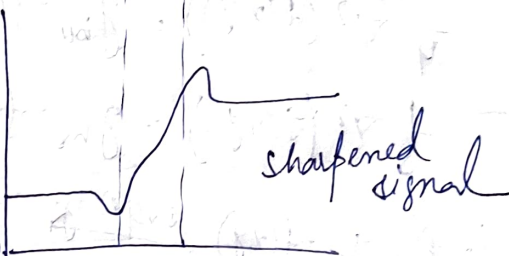
$$g(x, y) = f(x, y) + k g_{\text{mask}}(x, y)$$

$k \geq 0$

when $k = 1 \Rightarrow$ unsharp masking.

when $k > 1 \Rightarrow$ highboost filtering.

when $k < 1 \Rightarrow$ reduces the contribution of the unsharp mask.



Homomorphic filtering

An image can be expressed as the product of its illumination $i(x, y)$ and reflectance $r(x, y)$ as —

$$f(x, y) = i(x, y) \cdot r(x, y)$$

$$z(x, y) = \log(f(x, y)) = \log i(x, y) + \log r(x, y)$$

$$\begin{aligned} \text{Then, } F[z(x, y)] &= F[\log f(x, y)] \\ &= F[\log i(x, y)] + F[\log r(x, y)] \end{aligned}$$

$$\Rightarrow Z(u, v) = F_i(u, v) + F_r(u, v)$$

"
We can filter $Z(u, v)$ using a filter transfer function $H(u, v)$

so that,

$$S(u, v) = H(u, v) Z(u, v)$$

$$= H(u, v) F_i(u, v) + H(u, v) F_r(u, v)$$

The filtered image in the spatial domain is then —

$$s(x, y) = F^{-1}[S(u, v)]$$

$$= F^{-1}[H(u, v) F_i(u, v)] + F^{-1}[H(u, v) F_r(u, v)]$$

$$= i'(x, y) + r'(x, y)$$

$$g(x, y) = e^{s(x, y)}$$

$$= e^{i'(x, y)} \cdot e^{r'(x, y)}$$

$$= i_0(x, y) \cdot r_0(x, y)$$

This method is based on a special case of a class of systems known as homomorphic systems. The key to approach is the separation of the illumination and reflectance components achieved. The homomorphic filter transfer function, $H(u, v)$, then can operate on these components separately.

$$f(x, y) \rightarrow \boxed{\log} \rightarrow \boxed{\text{DFT}} \rightarrow \boxed{U(u, v)} \rightarrow \boxed{\text{IDFT}} \rightarrow \boxed{\exp} \rightarrow g(x, y)$$

$$H(u, v) = (\gamma_H - \gamma_L) \left[1 - e^{-\frac{c D^2(u, v)}{D_0^2}} \right] + \gamma_L$$

constant c controls the sharpness of the slope of the function as it transitions between γ_L and γ_H

